Detecting Macro Regimes with a Hidden Markov Model

Sebastian Wright · Finance, University of Florida (Warrington) ·
github.com/SebastianIWright/hmm-macro-regime-detection

The Question

Asset classes behave differently across the business cycle, but nobody announces the regime in real time. Can a statistical model spot it from macro data alone, and does knowing it improve a portfolio once you remove look-ahead bias?

Approach

- Pulled 8 FRED macro series (GDP, CPI, unemployment, manufacturing jobs, 10Y-2Y spread, Fed Funds, HY spread, sentiment) and monthly returns for SPY, TLT, GLD, and GSG.
- Engineered momentum features and compressed 11 inputs to 4 PCA factors (81% of variance).
- Fit a 4-state Gaussian HMM (10 restarts) and mapped states to Expansion, Stagflation, Recession, and Recovery.
- Gave each regime target weights, backtested net of 10bps costs, then re-ran it walk-forward, retraining every 6 months on only past data.

Results

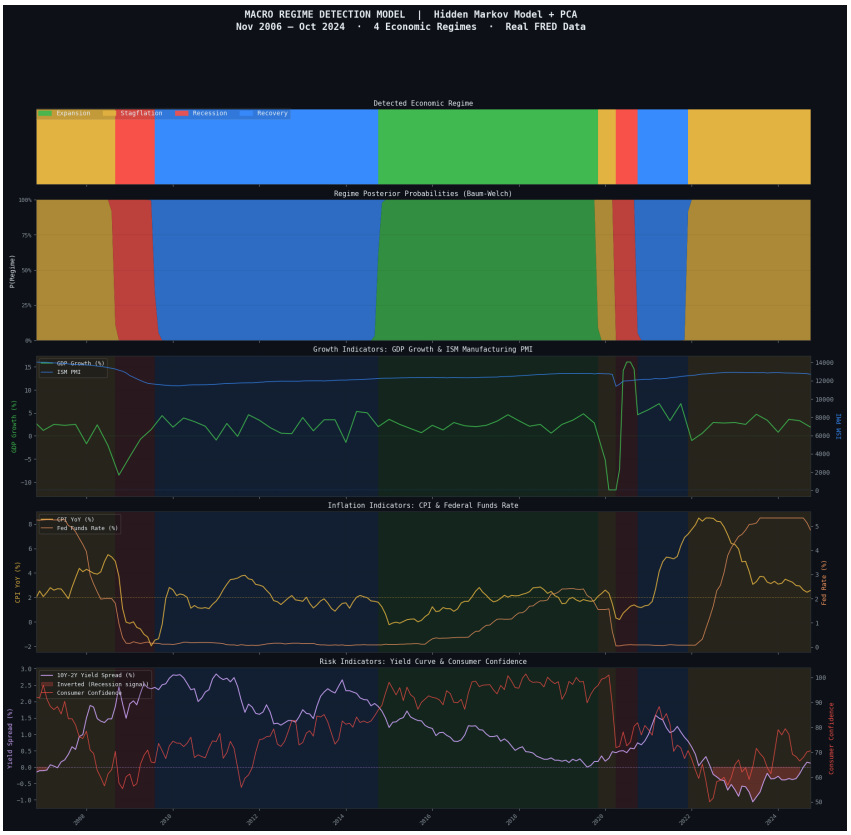
Metric	Regime-Aware	60/40
Full sample ann. return	10.1%	8.0%
Full sample Sharpe	0.50	0.31
Full sample max drawdown	-19.5%	-28.5%
Walk-forward ann. return	5.3%	7.7%
Walk-forward Sharpe	0.55	0.71
Walk-forward max drawdown	-16.9%	-26.2%

Takeaways

- In-sample, the model looked like it added 2% a year. Walk-forward, the return edge disappeared. That gap is the cost of look-ahead bias.
- What held up out of sample was risk control: about 9 points less drawdown than 60/40.
- Weights set by economic logic beat weights learned from data. With only a few recessions in the sample, there isn't enough history to fit them reliably.

Next Steps

- Use regime probabilities to scale risk instead of switching allocations outright.
- Add more history and faster-moving market inputs, like credit spreads and the VIX.



Detected regimes over time.